

SENTRIX : Multimodal Edge Threat Intelligence System

Thapar Institute of Engineering & Technology | Computer Science & Engineering

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98.7%

Violence Recall

ResNet-18 Fight Model

100%

Acoustic Threat Acc

2D Mel-Spectrogram CNN

15.8 ms

Real-Time Latency

~63 FPS on Apple Silicon

 $R^2=0.984$

XGBoost Late Fusion

RMSE = 0.042 Error

₹3,240

Base Hardware BOM

ESP32-S3 (≤ ₹20k Limit)

110 dB

Physical Siren

Relay Deterrent Actuation

FOUNDATION

Problem & Architecture

- **94%-98% False Alarm Crisis:** Single-modality sensors trigger false alarms, costing \$2B+ annually in wasted dispatches (SIA 2022).
- **Indian Emergency Response:** 15-45 min latency; festive soundscapes fool audio mics.
- **The SENTRIX Solution:**
 1. Multimodal Late Fusion (Vision + Audio + Face).
 2. 100% Zero-Cloud Edge Autonomy.
 3. Tamper-Proof AES-256-GCM Evidence Vault.

CORE AI

Multimodal AI & TCI Engine

Threat Confidence Index (TCI: 0.00 - 1.00) Formula

$$TCI = \text{XGBoost}(S_{\text{vis}}, S_{\text{aud}}, S_{\text{mot}}, S_{\text{beh}}, S_{\text{id}}, S_{\text{weap}}, S_{\text{fire}}, I_{\text{night}})$$

FOUR PARALLEL DEEP NEURAL MODELS:

- YOLOv8n Weapons & Fire: 0.884 mAP | 5.05 ms
- ResNet-18 Fight Classifier: 98.7% Acc | 4.80 ms
- 2D Mel Audio CNN: 100.0% Validation Acc | 1.20 ms
- Face ReID Engine: 128-d Metric Cosine Match

5-TIER AUTONOMOUS ESCALATION MATRIX:

Level 1 (0.00-0.20) : SECURE

Passive monitoring, background buffer sync

Level 2 (0.20-0.40) : SUSPICIOUS

Soft UI warning, frame lock, event DB creation

Level 3 (0.40-0.65) : ELEVATED

Amber HUD alert, AES-256 encrypted snapshot

Level 4 (0.65-0.85) : HIGH THREAT

Emergency dispatch prep, encrypted video clip

Level 5 (0.85-1.00) : CRITICAL

110dB Siren Relay ON + Instant Police Dispatch

RESULTS

Validation & Compliance

EMPIRICAL BENCHMARK SCORECARD:

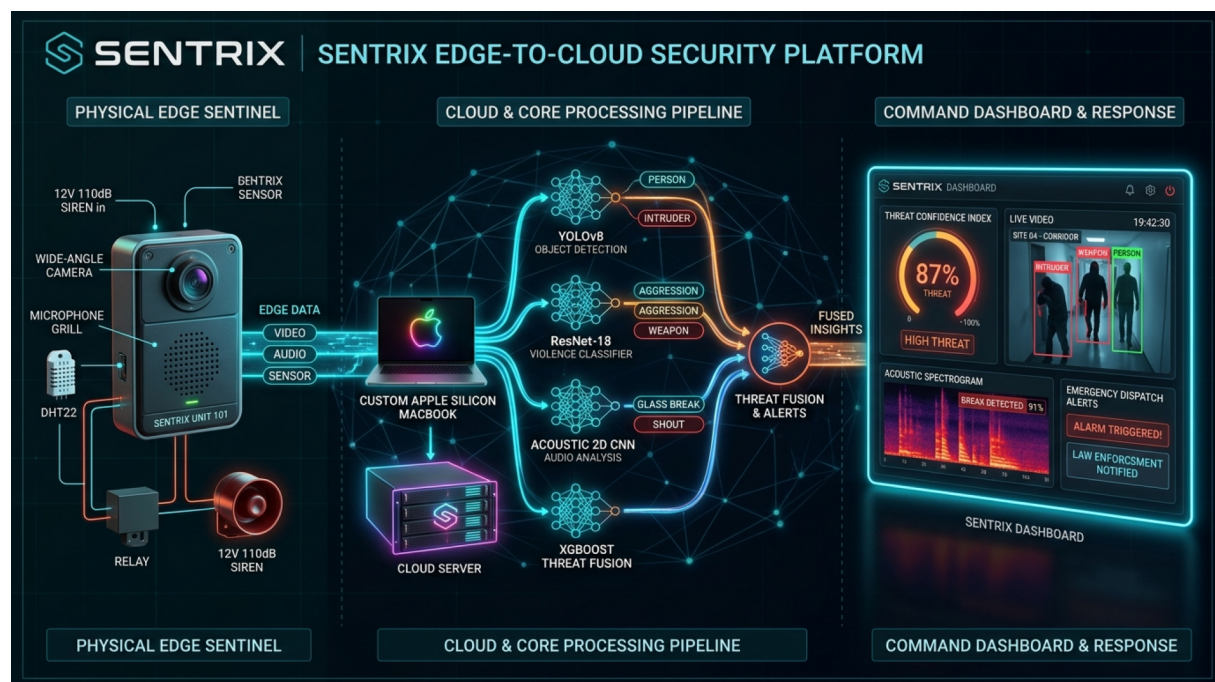
- Violence Model: 98.7% Train / 82.6% Val (19.6k frames)
- Acoustic Threat CNN: 100.0% Validation Accuracy
- XGBoost Late Fusion: $R^2 = 0.984$ (RMSE = 0.042)
- Real-Time Latency: 15.83 ms (~63 FPS on Apple Silicon)

CRYPTOGRAPHIC EVIDENCE VAULT:

- Authenticated Encryption: AES-256-GCM Sealing
- Forensic Integrity: SHA-256 Sidecar Fingerprint
- Court Admissibility: Guaranteed chain-of-custody

STATUTORY & REGULATORY STANDARDS:

- BIS IS 16910 / IEC 62676 Video Surveillance Standard
- MoHUA Smart Cities Mission Video Analytics Guidelines
- CERT-In Cybersecurity Directions for Forensic Data
- OSHA 1910.95 / NFPA 72 Acoustic Alarm Specifications



DUAL HARDWARE BOM (≤ ₹20,000 INR):

- Arch A (ESP32-S3 Wi-Fi Node): ₹3,240 INR
- Arch B (Orange Pi 5 SBC Node): ₹14,850 INR